Mock BE Practical Viva - Complete Theory Notes

Mock BE Practical Viva - Complete Theory Notes

1) Feature Transformation - PCA (Principal Component Analysis)

PCA reduces high-dimensional data into fewer dimensions while retaining most variance. Converts correlated variables into uncorrelated ones (Principal Components). Steps: standardize data -> covariance matrix -> eigenvalues/vectors -> select top components -> transform data. Used for dimensionality reduction, e.g., distinguishing red vs white wine.

2) Regression Analysis - Uber Fare Prediction

Regression predicts continuous variables. Linear regression models relation between dependent (fare) and independent variables (distance, time). Ridge (L2) and Lasso (L1) prevent overfitting. Metrics: R2, RMSE. Steps: Preprocess -> detect outliers -> check correlation -> train -> evaluate.

3) Classification Analysis - KNN

KNN is a supervised algorithm classifying data based on nearest neighbors using distance metrics. Steps: feature scaling -> choose K -> train-test split -> predict -> evaluate using confusion matrix, accuracy, precision, recall.

4) Clustering Analysis - K-Means

K-Means groups data into K clusters (unsupervised). Steps: choose K -> initialize centroids -> assign points -> update centroids -> repeat. Elbow method finds optimal K. Used on Iris dataset for species grouping.

5) Ensemble Learning - Random Forest, AdaBoost, GBM, XGBoost

Ensemble combines models to improve accuracy. Random Forest = multiple decision trees (bagging). Boosting methods (AdaBoost, GBM, XGBoost) build sequential trees minimizing errors. Voting combines predictions. Metrics: Accuracy, F1-score, ROC-AUC.

6) Reinforcement Learning (Maze/Taxi/Tic-Tac-Toe)

RL trains an agent to act via rewards and penalties. Agent learns optimal policy using states, actions, rewards. Q-learning updates Q-values to maximize cumulative reward. Examples: Taxi-v3, Tic-Tac-Toe game.

7) Data Cleaning and Preparation - Telecom Churn

Ensures accurate, consistent data. Steps: import -> handle missing values -> remove duplicates -> fix inconsistencies -> convert data types -> detect outliers -> feature engineering -> scale data -> split for modeling.

8) Data Wrangling - Real Estate Market

Transforms raw data for analysis. Steps: import and rename -> handle missing/outliers -> merge -> filter -> encode categories -> aggregate (avg price by area). Goal: prepare clean data for modeling and insight.

9) Data Visualization - Air Quality Index

Graphical data representation using matplotlib. Use line plots (AQI trend), bar plots (pollutant levels), box plots (distribution), scatter (AQI vs pollutant). Adds labels, legends, titles. Goal: find trends and pollution patterns.

10) Data Aggregation - Retail Sales

Groups and summarizes data. Steps: group by region/category -> compute totals -> visualize (bar/pie/stacked bar) -> identify top regions. Goal: analyze regional sales performance.

Viva Tips:

- Define -> explain -> give example.
- Revise metrics: Accuracy, Precision, Recall, F1, R2, RMSE.
- Understand supervised, unsupervised, and reinforcement learning differences.
- Stress importance of data preprocessing and visualization.